

MathMind AI — RAG Test Document

1. Algebra

An algebraic expression combines numbers, variables, and mathematical operations. A linear equation is an equation in which the highest power of the variable is one. For example, $2x + 5 = 15$ is a linear equation. Solving it gives $x = 5$.

2. Quadratic Equations

A quadratic equation has the general form $ax^2 + bx + c = 0$, where a is not equal to zero. The discriminant is $D = b^2 - 4ac$. If D is positive, there are two distinct real roots. If D is zero, there is one repeated real root. If D is negative, there are no real roots.

3. Matrices

A matrix is a rectangular arrangement of numbers in rows and columns. For a 2 by 2 matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, its determinant is $\det(A) = ad - bc$. For example, the determinant of $\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$ is $2(4) - 3(1) = 5$.

4. Calculus

Differentiation measures the rate at which a function changes. For the function $f(x) = x^2$, the derivative is $f'(x) = 2x$. Integration can be viewed as the reverse operation of differentiation. For example, the integral of $2x$ with respect to x is $x^2 + C$.

5. RAG Test Facts

Question	Expected fact
What is a linear equation?	An equation whose highest variable power is one.
What is the discriminant?	$D = b^2 - 4ac$.
When is a quadratic's discriminant zero?	When the quadratic has one repeated real root.
What is the determinant of a 2x2 matrix?	$ad - bc$.
What is the derivative of x^2 ?	$2x$.
What is the integral of $2x$?	$x^2 + C$.